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KHALĪFAH • AMĀNAH • IQRA' • RAHMATAN LIL-ĀLĀMĪN

MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)
SEM 2 2025/2026

LAB 07:
INTERFACING A PLC WITH A MICROCONTROLLER AND A PC OVER
ETHERNET/IP

SECTION 1
GROUP 10

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Abstract

This experiment focuses on developing and implementing a Programmable Logic Controller (PLC) system interfaced with an Arduino microcontroller and a PC through Ethernet/IP communication using OpenPLC software. The main objective was to understand the principles of PLC programming, ladder diagram development, and hardware interfacing in industrial automation applications. Initially, a simple blinking LED circuit was designed to demonstrate basic PLC operations, variable assignment, and hardware interfacing. The experiment was then extended to a Start-Stop Control Circuit using push buttons and LED indicators, simulating real industrial motor control applications. The ladder diagrams were designed and simulated in OpenPLC Editor before being compiled and transferred to the Arduino board. Proper pin configuration and communication setup were carried out to ensure successful operation of the system. The ladder diagrams were designed and simulated in OpenPLC Editor before being compiled and transferred to the Arduino board. Proper pin configuration and communication setup were carried out to ensure successful operation of the system. This experiment highlights the importance of PLC systems in industrial automation, the practical use of Ethernet/IP communication for device integration, and the effectiveness of combining PLCs with microcontrollers for cost-effective automation solutions.

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Introduction

Programmable Logic Controllers (PLCs) are specialized digital computers widely used in industrial automation to control machinery, manufacturing processes, and electromechanical systems with high reliability and efficiency. Compared to conventional relay-based control systems, PLCs provide greater flexibility, easier troubleshooting, faster operation, and simpler modification through software programming. One of the most common programming languages for PLCs is Ladder Diagram (LD), which visually represents control logic in a form similar to electrical relay circuits, making it intuitive for industrial applications.

PLC systems can now be integrated with microcontrollers and computers through Ethernet/IP networks to improve system monitoring, data exchange, and remote operation as the industrial communication technologies now are advanced. Ethernet/IP communication allows multiple devices to communicate efficiently within the same network. In this experiment, OpenPLC Editor software is used to design, simulate, and transfer ladder logic programs to an Arduino microcontroller, enabling the Arduino to function as a PLC-based control device.

The experiment starts with developing a simple blinking LED circuit to help students understand the basic concepts of ladder logic programming, variable configuration, and hardware interfacing. Then, it continued with a Start-Stop Control Circuit, which is commonly applied in industrial motor control systems. This circuit demonstrates essential PLC concepts, including latching operations, push-button input control, and output holding functions. The ladder diagrams were designed and simulated using OpenPLC Editor before being compiled and uploaded to the Arduino board for practical implementation. Overall, a deeper understanding of PLC programming, Ethernet/IP communication, and the integration of software and hardware within industrial automation systems can be developed from this experiment.

2. Material Equipment

- Laptop
- OpenPLC Editor software
- Arduino Uno
- Breadboard
- 2 Push button switches (Start and Stop buttons)
- LED
- Resistors (220 Ω for LED and suitable pull-down/pull-up resistors for switches)
- Jumper wires
- USB cable for Arduino connection

3. Experimental Setup

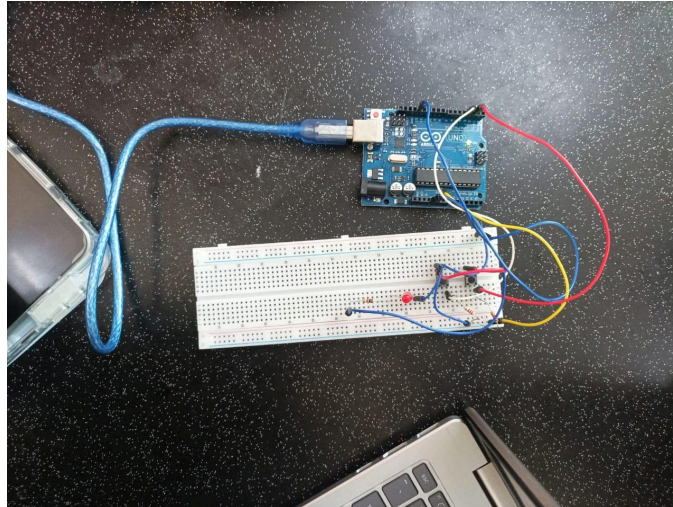


Figure 3.1 Circuit Setup Diagram

The experiment was carried out by interfacing the ladder logic program created in OpenPLC Editor with an Arduino board. The Start and Stop push buttons were connected to the digital input pins of the Arduino, while the LED was connected to a digital output pin through a current-limiting resistor.

The ladder diagram was created in OpenPLC Editor based on the Start–Stop control logic. The program was compiled and simulated before being transferred into the Arduino board through the correct COM port. The OpenPLC variable addresses were mapped to the corresponding Arduino input and output pins.

The Start button acted as a normally open, push button to activate the output, while the Stop button acted as a normally closed, push button to deactivate the output. When the Start button was pressed, the LED remained ON through a holding contact because of the latch in the ladder logic. Pressing the Stop button interrupted the circuit and turned the LED OFF.

4. Methodology

1. OpenPLC Editor software was downloaded and launched.
2. A new ladder diagram project was created using the LD (Ladder Diagram) language.
3. Input and output variables were defined for the Start button, Stop button, and LED indicator.
4. The Start–Stop control ladder diagram was constructed according to the given circuit design.
5. Left and right power rails, contacts, coils, and holding logic were added into the ladder diagram.
6. The program was compiled to check for logic errors.
7. The ladder logic simulation was executed in OpenPLC Editor to verify proper operation.
8. The Arduino board was connected to the computer through a USB cable, and the COM port number was identified.
9. The OpenPLC variables were associated with the correct Arduino digital input and output pins.
10. The ladder diagram program was transferred to the Arduino board.
11. The hardware circuit was assembled on a breadboard using push buttons, LED, resistors, and jumper wires.
12. Functional testing was carried out by pressing the Start and Stop buttons to observe the LED operation.

5. Data Collection

The pushbutton-based Start-Stop Control Circuit, which was created using OpenPLC and Arduino, was observed in order to gather data. The LED output's reaction to various pushbutton conditions was noted throughout the experiment. The LED was turned on by pressing the Start pushbutton and turned off by pressing the Stop pushbutton. To make sure the system functioned reliably and precisely, a number of tests were carried out. Both pushbutton input signals and the matching LED output conditions were manually observed and documented. Additionally, before transferring the program to the Arduino board, the ladder diagram simulation in OpenPLC Editor was observed to confirm that the logic sequence operated as intended. The COM port connection, successful compilation status, and hardware response during testing were also observed as part of the experimental data collection. The collected data mainly focused on the functionality, response accuracy, and stability of the control system.

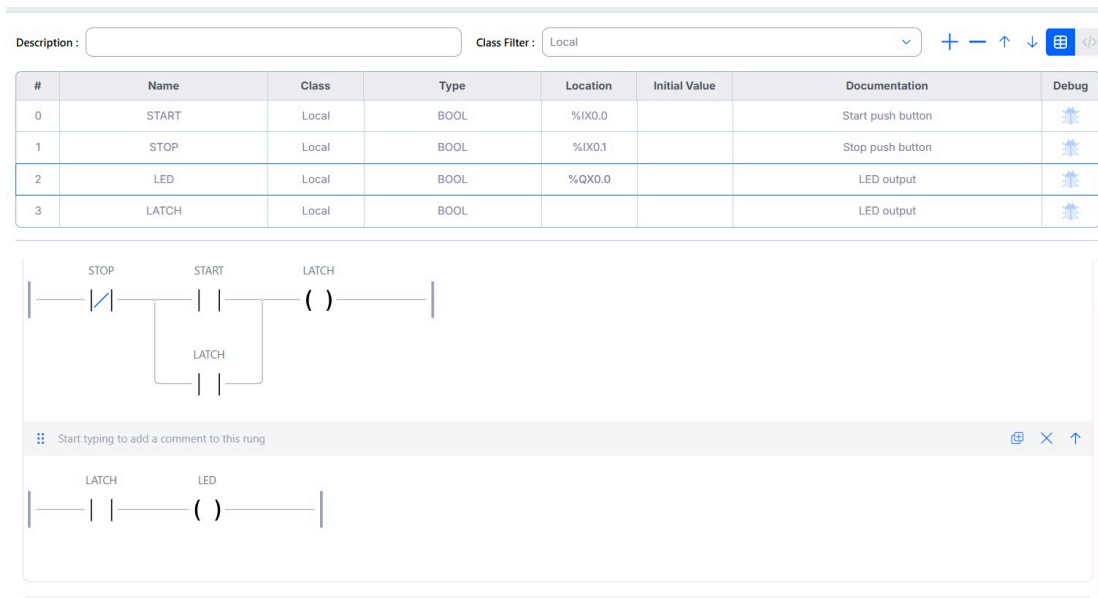


Figure 5.1- Ladder diagram used for Start-Stop control circuit

6. Data Analysis

By contrasting the LED output response with the pushbutton input conditions, the gathered data was analyzed. The ladder diagram's latch instruction triggered the LED when the Start pushbutton was pressed, and it remained in the ON state even after the pushbutton was released. This shows that the latch logic worked properly. The latch circuit was then disrupted when the Stop pushbutton was pressed, instantly turning off the LED. Repeated testing demonstrated that there was no discernible delay or system error in the Arduino and OpenPLC communication. The actual hardware output on the breadboard and the simulation results in OpenPLC Editor were also contrasted. Similar operational behavior was produced by both simulation and hardware testing, demonstrating that the ladder diagram logic was appropriately transferred to the Arduino board. The analysis verified that the designed ladder logic sequence was followed by the Start-Stop Control Circuit.

7. Results

The experiment was successfully completed, and the pushbutton-based Start-Stop Control Circuit operated as expected. The Start pushbutton successfully activated the LED output, while the Stop pushbutton deactivated the LED output. The latch instruction allowed the LED to remain ON continuously until the Stop button was pressed.

Conclusion

In conclusion, the experiment was successfully conducted by developing and testing a pushbutton-based Start-Stop Control Circuit using OpenPLC and Arduino. The system was able to perform according to the designed ladder logic, where the Start pushbutton activated the LED output and the Stop pushbutton deactivated it correctly.

The experiment also showed that the latch instruction worked effectively in maintaining the LED condition until the Stop button was pressed. In addition, the OpenPLC simulation and the actual hardware testing produced consistent results, proving that the program was transferred successfully to the Arduino board.

Overall, the experiment provided a better understanding of PLC ladder logic programming, pushbutton control systems, and the integration between OpenPLC software and Arduino hardware. The developed system can be applied in simple industrial automation applications such as motor Start-Stop control systems.

Discussion

The experiment clearly demonstrated how to use an Ethernet/IP-based, OpenPLC implementation to connect a PLC ladder logic program with a microcontroller. The desired ladder logic structure was followed by the Start-Stop control circuit.

When the Start push button was pressed, the output coil was energized and the LED turned ON. The holding contact in the ladder diagram maintained the output state even after the Start button was released. When the Stop push button was pressed, the circuit path was interrupted, causing the output coil to de-energize and the LED to turn OFF. This behavior confirmed the correct implementation of latching logic in the ladder diagram.

The simulation feature in OpenPLC Editor was useful for identifying programming errors before transferring the program to the Arduino hardware. Correct assignment of OpenPLC addresses to Arduino pins was important to ensure proper communication between the software and hardware. Incorrect COM port selection or wrong pin mapping could cause upload failure or improper circuit operation.

Recommendation

To improve the effectiveness and reliability of this experiment, several recommendations can be implemented. First, the pin configuration and COM port settings should be verified carefully before uploading the ladder diagram to the Arduino board in order to prevent communication errors between the software and hardware. Then, check all circuit connections properly to ensure stable operation of the Start-Stop Control Circuit and to minimize the risk of hardware malfunction. Lastly, Human Machine Interface (HMI) monitoring or Internet of Things (IoT) communication features for remote monitoring and control should be implemented as it allows us to gain a better understanding of modern Industry 4.0 technologies and smart manufacturing systems.

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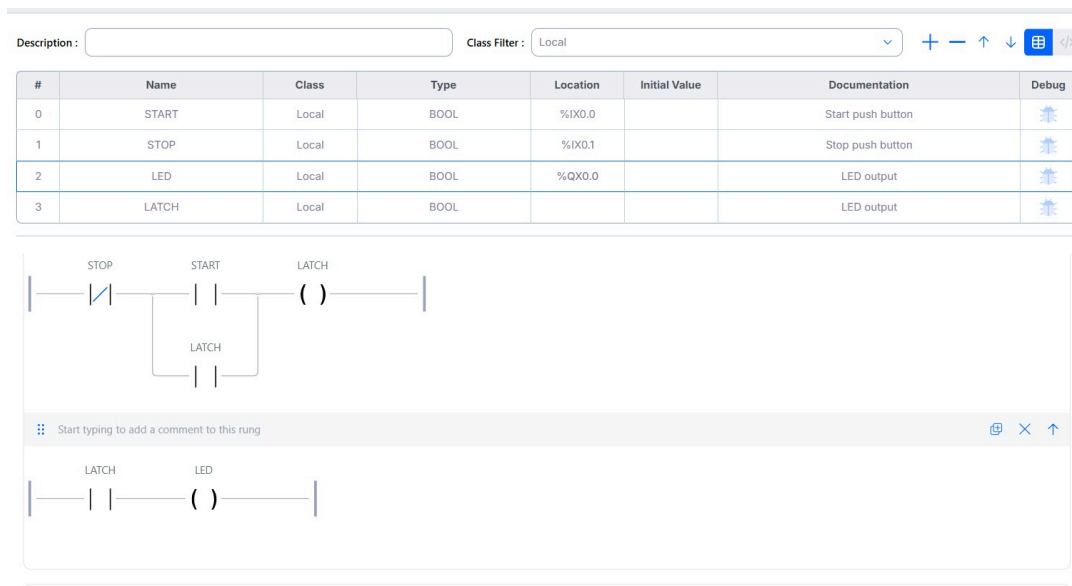
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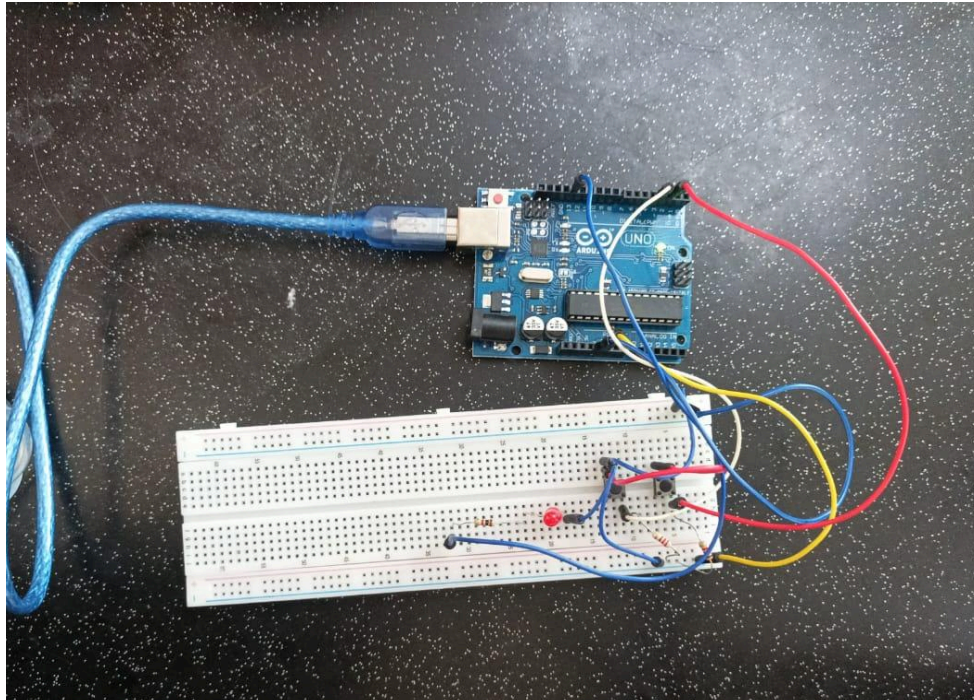
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Appendices



Appendices 1: Ladder Diagram in OpenPLC Editor



Appendices 2: Circuit setup

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We would also like to thank all our group members for their cooperation and teamwork. Everyone contributed their ideas and effort, which made it easier for us to complete all the tasks and overcome the challenges during the lab sessions.

Besides that, we are grateful to our classmates for their help and discussions, which made the learning process more enjoyable and meaningful.

Student's Declaration

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

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Contribution: Abstract, Introduction, , Recommendation, References, Appendices and Acknowledgement.